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SCIENCE

ABOVE AND BEYOND DRIVING ABILITIES: TOWARD A SINGLE INDEX

The present study allows us to pave the way toward a single and common indicator of driving behavior. We investigated performance in several tests known to assess key process of driving fitness such as driving maneuvering, visual and visuo-cognitive abilities. We propose a methodology to bring together all these measures under a single index we call the Driver's Safety Index. The results indicate that this index might be helpful to characterize driving behaviors and to pin down the kind of limitations an individual must deal with to effectively control a vehicle. The present investigation is the first step toward an intended index, which might be translated in licensing policies. It also calls for further clinical investigations to refine the visual criteria necessary in its computation. Importantly, we evidence that visuo-cognitive performance, a parameter overlooked up to now, is crucial to consider for assessing driving abilities and should be incorporated in such an index.



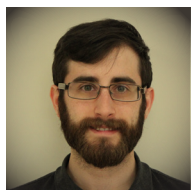
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Based on the understanding of the vision of human in mobility, and how it perceives, makes decision and interacts with our world in movement, the NSERC-Essilor VisAge Chair has an impressive story from 2001 until now, fostering innovation in the field of optics and vision health. **Here, we focus on one of its recent exploration and understanding: how to define driving behavior.**

The industrialized world is continuing to see a marked demographic transformation, with the proportion of individuals over the age of 65 growing ever higher. This demographic shift and its inevitable social, medical and economic consequences has been and will continue to represent a tremendously important topic of investigation for driving safety. Indeed, when accounting for the relative amounts of kilometers they drive, elderly drivers are known to be involved in more fatal crashes (Insurance Institute for Highway Safety, 2014)¹ and have more traffic convictions when compared to any other adult age group. Considering that driving is the principal mode of travel for adults worldwide it is crucial to understand what factors are involved in the observed age-related decline in driving fitness to translate these factors in licensing policies.

In the few countries assessing more than simple vehicle maneuvering abilities, the second criteria considered to obtain or renew a driving license is visual acuity. This is undoubtedly consistent since it

KEYWORDS

driving, vision, visuo-cognitive abilities, index, safety

Table 1: Description of the nine driving measures involved

Measures	Code	Description
Crash	D1	Whether collisions occurred or not
Near Crash	D2	When a collision was avoided at the very last moment
SDLP	D3	Standard deviation of lateral position
Max Brake	D4	Hardest amount of braking
Dist. at Max Brake	D5	Distance at which « Max Brake » is recorded
Max Steer Change Rate	D6	Most extreme (in terms of range and speed) left or right steering wheel position change
Dist. At Max Steer Rate	D7	Distance at which « Max Steer Change Rate » is recorded
Steer Range	D8	Difference in degrees between leftmost and rightmost steering wheel position
Mean Driving Speed	D9	Naturally adopted speed

is well-accepted that good eyesight is preponderant to drive safely (see for a review Owsley & McGwin, 2010).² **Indeed, vision is a crucial sense to make decisions on the road, and poor vision can notably decrease chances of reacting in time, putting oneself and others at greater risk.** Nevertheless, eyesight is only the first step in perception mechanisms and how we use this visual information, whether or not it is altered by age-related ocular diseases, is also a key parameter in the decision-making process when facing or anticipating an impending hazard. This is especially true in a daily task such as driving which, despite its apparent ease, is known to be a complex task involving multiple visuo-cognitive processes such as visuo-spatial skills, processing speed as well as attentional processes. Accordingly, we think that simple measures of vehicle manoeuvring and visual acuity alone cannot be sufficient to faithfully capture the whole and complex picture of driving fitness.

In the context of the VisAge chair, we seek to better define the different drivers' profiles to be able to offer them individualized solutions. **Therefore, the present study aimed at better characterizing drivers' profiles through i) a further exploration of these visuo-cognitive abilities and ii) linking them with the other determining factors such as visual abilities and vehicle maneuvering measures.**

Methodology

A total of 115 licensed drivers between the ages of 18 and 86 were recruited for the study. To assess the efficiency of our characterization among a wide age-range, the participants were separated into two groups: 64 young participants (mean age= 28.8 ± 10.23 (SD) years old) and 51 older participants (mean age= 77.2 ± 5.01 (SD) years old). Each participant was involved in three different experimental phases:

1) Visual tests including a visual acuity test (Snellen chart), a stereoscopic vision test (Randot test) and a binocular Esterman visual field test run on the Humphrey Visual Field Analyzer. Those tests gave rise to an accurate score representing a visual ability subclass: V1-Acuity, V2-Stereo and V3-Visual Field, respectively.

2) Driving tests were performed using a high-fidelity, moving base driving simulator (Figure 1). Three scenarios were designed to represent natural driving environments with an increasing visual attentional load: highway (i.e. low), rural (i.e. middle) and city (i.e. high). Based on a recent method developed at the NSERC-Essilor Chair, the rural scenario appeared as the most efficient for detecting subtle differences in driving maneuver ability (Michaels et al. 2017).³ We also identified some variables merging as significant and non-redundant measures of driving behavior (see Table 1). Since we aimed at characterizing drivers' behaviour, the driving measures presented hereinafter will be those recorded during the rural scenario.

3) A visuo-cognitive task known as 3-Dimensional Multiple Object Tracking (3D-MOT; Figure 2) to assess the individual's ability to capture and integrate relevant information in a highly complex visual environment (Pylyshyn & Storm, 1988; Faubert & Sidebottom, 2012).^{4,5} The individual's score (hereinafter called "visuo-cog") corresponds to the mean speed threshold at which he was able to simultaneously track 4 among 8 balls. To further explore the relevance of the visuo-cognitive measure, we investigated the link between this measure and the usual driving and visual measures mentioned above.

Main findings

As illustrated in Figure 3 panel left, the results evidenced strong correlations between the visuo-cognitive score and several driving measures as well as with some visual measures. Visuo-cognitive score is correlated with driving measures known to be main indicators of diminished



« Taken together, these results reinforce the idea that visuo-cognitive abilities are key processes in car driving. »

driving abilities such as the number of crashes ($r = -.31$; $p < .001$), the mean speed naturally adopted ($r = .47$; $p < .001$) and the standard deviation of lane position ($r = -.26$; $p = .005$). Furthermore, this score is also correlated with the stereoscopic vision capacity ($r = -.44$; $p = .001$) and to a lesser extent with visual acuity ($r = -.25$; $p = .07$). Because one of the advantages of this visuo-cognitive task is the involvement of a three-dimensional virtual environment, the link between visuo-cognitive score and stereoscopic vision we observed was expected.

Additionally, the multiple linear regression analyses presented below reveal that the visuo-cognitive score is a better predictor of car driving and visual abilities as compared to chronological age and naturally adopted mean driving speed. Taken together, these results reinforce the idea that visuo-cognitive abilities are key processes in car driving. **As a relevant predictor of driving performances across age, this visuo-cognitive score should be included in the battery of visual tests for identifying drivers with hampered driving abilities.**

Integrative approach toward a single Driver's Safety Index

We established a methodology to link these three categories of driving behavior assessment by building a unique indicator. Determining a unique indicator of safe driving behavior represents both a need and a risk. This is needed to build screening tests, which could then be readily translated into licensing policies. Nevertheless, this is also a risk because resuming driving behavior into one unique score might lead to neglect or underestimation of important components of driving abilities. Instead of looking for the best predictor of driving behaviour, we proposed a global score in which each important driving component makes its contribution. Attempting to do so, we computed the area of the three different parts of a radar chart visualization and built a global index. The first area included the indicators from D1 to D9 and corresponds to an index of driving maneuver abilities ("Idri"). The second area involved the V1-Acuity, V2-Stereo and



Figure 1: Picture of the VS500M driving simulator.³

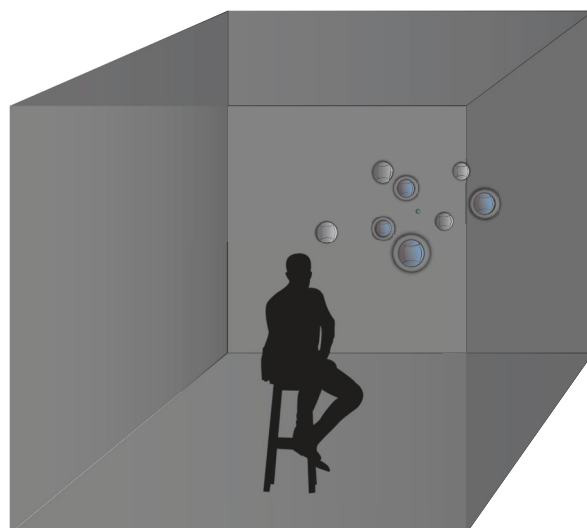


Figure 2: illustration of the visuo-cognitive task. Firstly, 8 randomly positioned spheres are presented in a virtual volumetric space. Secondly, the 4 spheres to be tracked during the trial are quickly highlighted in color. Thirdly, all spheres turn back to their original color and begin to move. Finally, the observer is asked to respond by identifying the spheres he had to track. Then, feedback is given to the observer. If the observer correctly identifies all spheres, the task is repeated at a faster speed. If, on the other hand, the observer makes a mistake, the task is repeated at a slower speed.

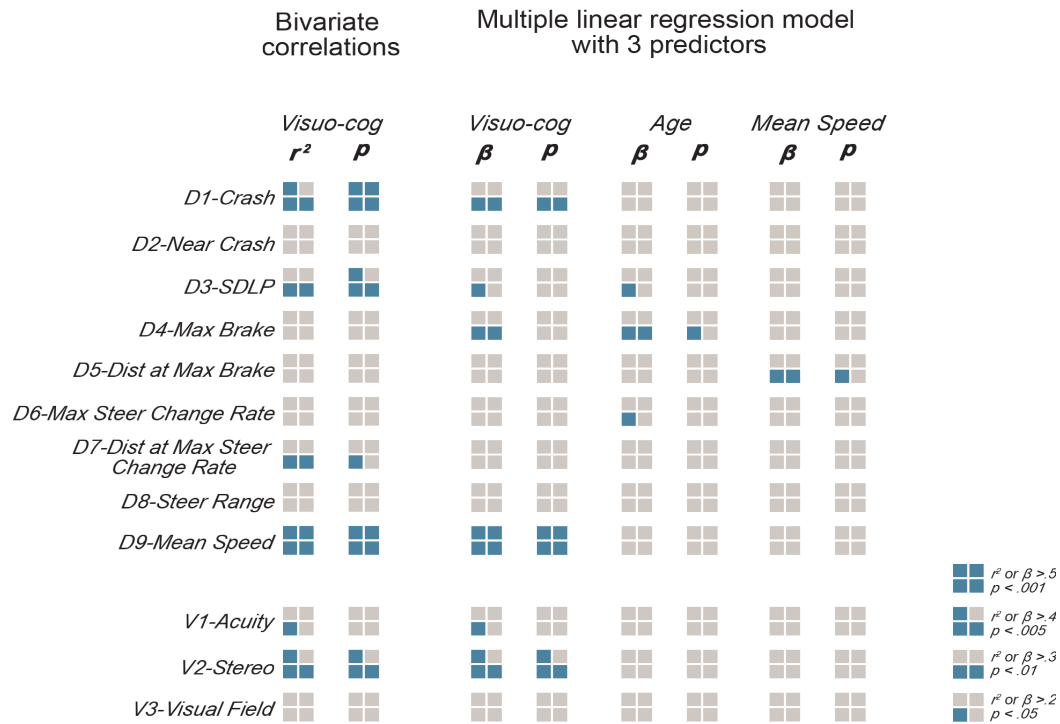


Figure 3: Bi-variate correlations and multiple linear regression analysis showing the relevance of the visuo-cognitive measure.

V3-Visual Field indicators and corresponds to an index of visual abilities ("Ivis"). The third area represents an index of visuo-cognitive abilities ("IvisCog"). Finally, the sum of these three areas gives rise to a global index, we called Driver's Safety Index (DSI), thought to represent the amount of limitations an individual must deal with to effectively control a vehicle.

Application of the Driver's Safety Index to a comparison between two individuals

The use of our DSI associated with a radar chart visualization show, in one straightforward and clear picture, which components of driving might be less developed in one individual and simplifies the comparisons between two individuals (Figure 4) or two groups (Figure 5).

As an example, when comparing the results obtained by the individuals showing the lowest and the highest visuo-cognitive scores in older (Figure 4a) as well as in younger group (Figure 4b), the radar chart visualizations and DSI appear to be informative and help to avoid misinterpretation. Indeed, if we only consider visual abilities, the conclusion we would draw would be that the two drivers have the same driving abilities. Nevertheless, considering DSI indicates that in both groups, individuals who obtained the best visuo-cognitive scores have more preserved driving skills than the drivers who obtained the lowest visuo-cognitive performance. This wide difference is revealed by a DSI in the highest performers being twice as high as the DSI of the lowest performers.

Application of the Driver's Safety Index to a comparison between two groups

When considering the lowest and highest performance among a population, differences are obviously wide. Nevertheless, when attempting to compare the mean performance obtained in two groups, differences might be more subtle and less straightforward to analyze. The picture below illustrates performances recorded for the oldest half and for the youngest half of individuals in the Older group (Figure 5a), and those recorded for the lowest half versus highest half in visuo-cognitive performance among the Older group (Figure 5b). In such a situation, the DSI appears to be a relevant tool to compare the two groups since it allows one to decipher that driving performance was not dependent on chronological age (DSI: Oldest= 61.25; Youngest= 61.95; $p = .89$) but, rather, that participants were classified depending on their visuo-cognitive performances (DSI: Lowest_VCI= 51.82; Highest_VCI= 70.98; $p < .001$).

Discussion

Visual and visuo-cognitive assessment for driving should be considered as important as other road safety campaigns by authorities. Despite this fact, strict regulations to obtain or renew a driving license are still weak in most countries. One of the reasons explaining this absence of standards is the lack of homogeneous, simple and reliable criteria. Moreover, when a visual criteria is included in the requirements, like in North America and Australia, the tests are usually restricted to visual acuity and visual field assessments. Importantly, previous works already showed that visual acuity, when considered as an independent contributor to renewal

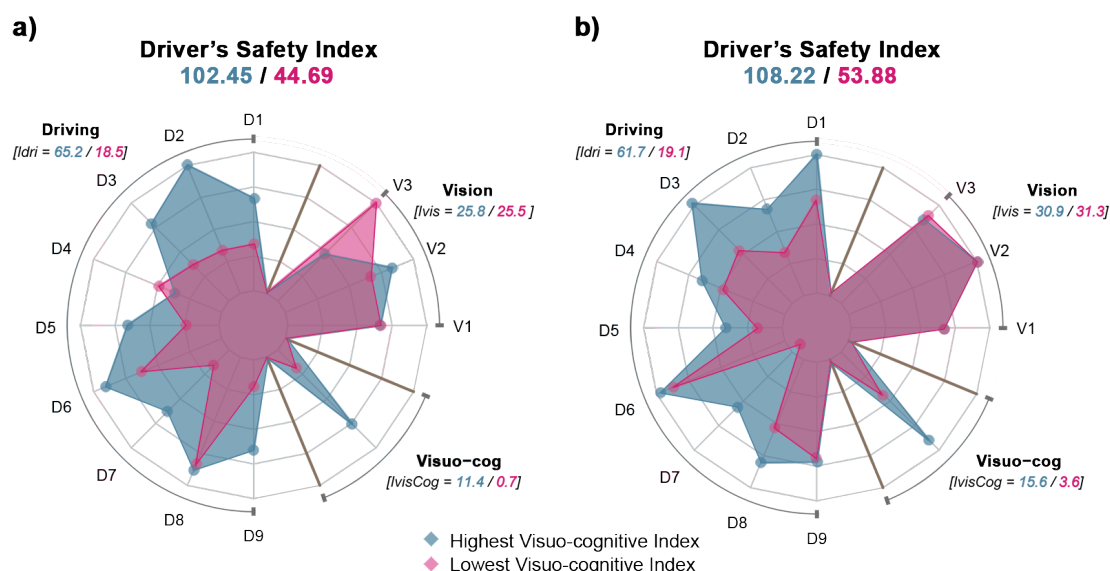


Figure 4: The radar chart representation and the global Driver's Safety Index (DSI). The individuals with highest (in blue) and lowest (in pink) visuo-cognitive index (VCI) among a) older and b) younger participants are represented. To balance / normalize the different indicators, all scores are expressed as a percentage of the best performance observed across the 115 participants and are thus reported on the same scale.

decision, had no impact on fatality rates in adults aged 65 years old (Grabowski et al. 2004)⁶ and no economic benefit for society (Viamonte et al. 2006).⁷

The two important results of this study help to highlight the relevance of the establishment or improvement of such criteria. Firstly, our results suggest that visuo-cognitive performance, as a central component of vehicle driving, can provide additional information on the participants' driving efficiency and the limitations they face. As such, it should be considered in the battery of tests leading to obtain or renew a driving license. Secondly, the computation of the Driving Safety Index might be helpful to easily characterize driving behavior with the advantage being that it merges a large panel of abilities that are recognized as central for safe driving.

Future investigations and improvements of the Driving Safety Index will bring new, more comprehensive knowledge of and a powerful method to characterize driving abilities. Here, we considered visual acuity, stereoscopic ability and the functional visual field of our participants, but many others criteria could be incorporated in its computation. Further studies like the ones from Wood & Owsley (2016)⁸, are needed to clarify which visual tests, such as for instance the contrast sensitivity test, are the most relevant to understand what visual deficits can hamper driving. Moreover, our observations on the driving simulator must be replicated in actual on-road driving. Finally, it is worth noting that we only included individuals with optimal or corrected-to-optimal vision. It will therefore also be of interest to assess the functionality of this indicator with a population suffering from visual deficits

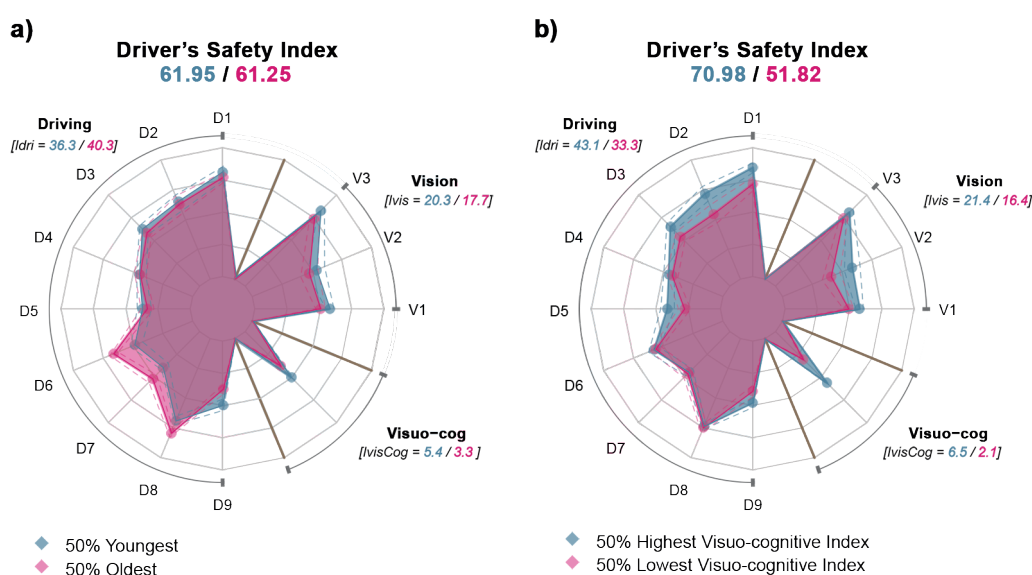


Figure 5: Characterization of two populations through Driver's Safety Index. a) Classification depending on the chronological age: comparison between the Oldest 50% (in pink) and the Youngest 50% (in blue) among individuals in the Older group. b) Classification depending on the visuo-cognitive indicator (VCI): comparison between the Lowest 50% (in pink) and the Highest 50% (in blue) visuo-cognitive scores among the older group. Dashed lines represent SEM.

Conclusion

We demonstrated that driving abilities, evaluated within a driving simulator, are strongly associated with the 3D-MOT scores. **This result highlights the importance to consider the visuo-cognitive abilities in the assessment of driving abilities.** Moreover, this study paves the way toward a single, common indicator of driving behaviour. However, its approval will require the involvement of eye care professionals and other health clinicians to further refine which visual criteria might be relevant to include in its computation. Refining these criteria will allow to frame them by standards data for each population or even targeting specific complains such as depth perception, glare or driving at night, and thus make the Driver's Safety Index more reliable and powerful. There is a two-fold challenge behind designing a measure upon which one can base decisions concerning license renewal: guaranteeing the optimum driving safety of road users while also preserving the autonomy of the elderly for as long as possible to avoid the potential adverse consequences of driving cessation.

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KEY TAKEAWAYS

- Assessing driving ability is a very difficult task and there has been no clear indicator until now.
- Two main needs are identified: to further explore an overlooked dynamic aspect of driving fitness (visuo-cognitive abilities) and to determine a unique indicator which could be readily translated into licensing policies.
- The results demonstrated that visuo-cognitive abilities should be considered by licensing authorities for identifying drivers with hampered driving abilities.
- We propose a global score, called Driver's Safety Index (DSI), in which each important driving component considered in the present study makes its contribution.
- The DSI is helpful in easily characterizing driving behavior with the advantage of merging a large panel of abilities that are recognized as central for a safe driving.
- The DSI is relevant to determine which components of driving are less developed in one individual and simplifies the comparisons between different individuals / groups.
- Our demonstration calls for further studies to clarify which visual tests would be the most relevant to unravel visual deficits that can hamper driving.
- The above takeaway suggests an increasing interest to involve eye care professionals in the improvement of this DSI.
- Behind the accuracy of this score there is a major public health issue: to guarantee the optimum driving safety of road users while preserving the autonomy of the elderly.